Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

1. Why is it important to avoid zeros in probability calculations when working with Markov models? 1 / 1 point

- The model becomes infinitely complex.
- Because they ruin the model's statistical balance.
- They make entire sequences possible, impacting predictions.
- They lead to impossible outcomes in the model.

Nice work

Correct! A zero probability excludes possible state transitions entirely.

2. What is a practical application of Markov models in Natural Language Processing (NLP)? 1 / 1 point

- Image recognition and processing
- Numeric prediction of stock prices
- Predictive text or autocomplete features in typing applications
- Speech to text transcription accuracy improvement

Nice work

Correct! They analyze sequences to predict likely following words.

3. What is the purpose of a train-test split in NLP? 1 / 1 point

- To ensure the model can generalize to new, unseen data
- To guarantee the trained model will perform perfectly
- To reduce the impact of noise in data
- To make data more manageable

Nice work

Correct! A train-test split helps verify that a model works well beyond its training data.

4. What strategy is commonly used to handle imbalanced classes in a dataset? 1 / 1 point

- Remove any data points from the larger class to balance.
- Exclusively use more complex models that handle imbalance.
- Employ resampling methods such as oversampling or undersampling.
- Assume the model will balance during learning automatically.

Nice work

Correct! Adjusting the class distributions helps balance the data.

5. What is stored in a second order transition dictionary for Markov models? 1 / 1 point

- Random choices made by current and past states.
- The first possible state only from any given state.
- The list of possible next states given two previous states.
- Only the probabilities of the current word state changes.

Nice work

Correct! The second order dictionary expands context based on two states of history.

6. What challenge has been noted with early attempts at using machine learning in content generation? 1 / 1 point

- Over-reliance on visual data
- Lack of coherence and contextual understanding
- Excessive computational speed
- Too focused on grammatical structure

Nice work

Correct! Early models often produced content that lacked contextual coherence.

7. What is a key characteristic of Markov models that allow them to be used in article spinning? 1 / 1 point

- Dependence on fully labeled datasets
- Ability to remember all previous states
- Ability to transition to the next state based purely on the current state
- Ability to generate infinite sequences

Nice work

Correct! This property allows Markov models to predict sequences in a probabilistic manner.

8. Why is automatically generating content from existing articles generally considered unethical? 1 / 1 point

- It provides better search engine optimization
- It enhances the original content
- It is universally accepted by content creators
- It often leads to plagiarism

Nice work

Correct! Article spinning can result in content that is not original and improperly attributed.

9. Why is tokenization an important process in article spinning? 1 / 1 point

- It makes articles shorter
- It converts articles into binary code
- It allows words to be replaced contextually
- It removes all punctuation from the article

Nice work

Correct! Tokenization breaks text into pieces that can be individually analyzed and manipulated.

10. What is a common limitation encountered by basic article spinning models when replacing text? 1 / 1 point

- Inability to understand long-range dependencies
- Inability to handle punctuation
- Inability to distinguish between synonyms
- Inability to process large text files

Nice work

Correct! Simple models struggle to maintain coherence across broader contexts.

11. Why is frequency analysis crucial in dealing with substitution ciphers? 1 / 1 point

- It provides randomness to prevent cracking
- It can predict exact message content without keys
- It simplifies the encryption process itself
- It helps identify likely letters in the cipher

Nice work

Correct! Frequency analysis identifies likely substitutions based on common patterns.

12. What is the main function of an encoding algorithm in a substitution cipher system? 1 / 1 point

- Decoding a received message
- Converting a plain message to cipher text
- Converting cipher text to plain text
- Finding randomness in messages

Nice work

Correct! This is the primary function of encoding.

13. How does randomness influence key generation in cryptography? 1 / 1 point

- It simplifies key retrieval
- It makes the key generation slower
- It adds predictability to the keys
- It prevents predictability in keys

Nice work

Correct! Randomness improves security by ensuring unpredictability.

14. How does changing one letter in a substitution cipher affect the message? 1 / 1 point

- It changes only the first instance of the letter
- It leaves the message unchanged
- It affects unrelated letters
- It modifies every occurrence of that letter

Nice work

Correct! Every occurrence of that letter in the text will be changed.

15. How do bigrams help in assessing decrypted messages? 1 / 1 point

- They simplify encoding
- They provide insight into possible sequences
- They define dictionary meanings
- They directly map to plain text

Nice work

Correct! Bigrams measure probability of character pairs lending insight into sequence accuracy.